Identifying and evaluating algebraic expressions

1 Match the following terms with their definitions:

a	Constant	**i**	Symbol that represents an unknown number
b	Variable	**ii**	A purely numeric term in an algebraic expression
c	Coefficient	**iii**	The value that indicates how many of a variable in a term
d	Algebraic term	**iv**	Term including a variable
e	Algebraic expression	**v**	A collection of variables, numbers, grouping symbols, and operations.

2 State the numerical coefficient of $-2.58t^4$.

3 Consider the expression $x + 6$.

a Find the value when $x = 4$.

b Find the value when $x = 9$.

c What happens to the value of $x + 6$ as we substitute in different values of x?

4 Evaluate the following expressions to one decimal place:

a $14.5 - 4x$ when $x = 4.2$

b $18.6 - 3x$ when $x = 4.1$

5 If $m = -3$ and $n = 4$, evaluate the following expressions:

a $mn - (m - n)$

b $m^2 + 9n$

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6 Evaluate the following expressions:

a $\dfrac{m}{2m}$ when $m = 5$

b $\dfrac{6}{7x + 3}$ when $x = 3$

c $\dfrac{a}{11} + \dfrac{b}{11}$ when $a = 5$ and $b = 2$

d $\dfrac{a}{1 - r}$ when $a = 3$ and $r = \dfrac{1}{10}$

e $\dfrac{p + 4}{q}$ when $p = -8$ and $q = -8$

f $\dfrac{3a}{2b - 6}$ when $a = 4$ and $b = 14$

g $\dfrac{2p + 4}{3q + 5}$ when $p = 2$ and $q = 4$

h $\dfrac{4r - 2}{3s - 5}$ when $r = 6$ and $s = 7$

i $\dfrac{x^2}{3} + \dfrac{y^3}{2}$ when $x = -4$ and $y = 3$

j $s\,(12 - t)$ when $s = -3$ and $t = 10$

k $x\left(2 - 4y^2\right)$ when $x = -2$ and $y = -2$

l $a\,(b - 13) + b^2$ when $a = -9$ and $b = 10$

m $3a^2 - 4b^3$ when $a = 4$ and $b = 2$

n $8p^2 + 5q^3$ when $p = 3$ and $q = -2$

o $5y - z^2 + 9$ when $y = 7$ and $z = 4$

p $\dfrac{1}{2}y^2 - 6z + 1$ when $y = -2$ and $z = \dfrac{5}{6}$

q $y^3 + 4yz - 2x^2$ when $x = -4$, $y = 2$ and $z = 1$

r $(u + v)\,(w - y)$ when $u = 5$, $v = 8$, $w = 2$ and $y = 10$

7 For $x = 5$ and $y = 4$, evaluate $\sqrt{2x^2 + 4y + 6}$ correct to two decimal places.

Additional questions

8 Write the expression $9x$ in words.

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9 State whether the following expressions are like terms.

a $11p$ and $3p$ **b** $2p$ and 15 **c** $9p$ and $6q$ **d** 12 and 8

e $5h$ and $5hk$ **f** $4h^4$ and $3h$ **g** $8h^3k^4$ and $9k^4h^3$ **h** $6h^k$ and $7k^h$

10 For the following algebraic expressions:

i State the number of terms.

ii Identify the numerical coefficient of the first term.

iii Identify the constant term.

iv State whether it contains like terms.

a $2x + 4$ **b** $7y + 3 + 5x$

c $3x + 2y - 8x + 9$ **d** $8p + 5$

11 Consider the algebraic expression $3x + 10 + 4x$.

a State the number of terms.

b Identify the numerical coefficient of the third term.

c Identify the constant term.

d State whether it contains like terms.

12 For each of the following, determine if the statement is true or false. Explain your reasoning.

a An expression must include a variable.

b Any expression with more than one term can be simplified.

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